

April 7, 2026

Dear Editors,

We hereby submit our manuscript “Magnetoelastic Waves in Ferromagnetic Thin Films Mediated by Dipolar Interactions” to *Applied Physics Express*.

Magnetoelastic coupling has attracted considerable attention from both fundamental and applied perspectives, particularly for the control and detection of magnetic states via mechanical degrees of freedom. In this manuscript, we present a theoretical formulation for the coupling between magnetostatic waves and elastic waves with micrometer-scale wavelengths, explicitly incorporating magnetic dipole-dipole interactions. At this length scale, dipole-dipole interactions play a significant role. However, their long-range nature makes them difficult to treat. Previous studies have therefore relied on approximate approaches that replace dipolar interactions with effective local terms, and a framework that directly treats dipole-dipole interactions has been lacking.

Here, we formulate the dipolar magnetic field by solving Maxwell’s equations using a Green’s function method in a thin-film geometry, taking into account elastic-wave-induced modulation of distances between magnetic dipoles. This enables us to derive the coupled equations of motion and quantitatively evaluate dispersion relations. As an example, we study a yttrium iron garnet (YIG) thin film and calculate the dispersion relation. Hybridization gaps of 0.1 to several MHz are found between magnetostatic and elastic (Lamb) waves, indicating clear mode hybridization. The coupling lies at the onset of the strong-coupling regime and is within experimental reach. Its magnitude is also consistent with reported magnetoelastic coupling constants in YIG, supporting the validity of our approach.

We believe this work provides a missing theoretical foundation for dipolar-mediated magnetoelastic coupling and will stimulate further studies in magnonics, phononics, and hybrid wave systems. We therefore expect it to be of interest to the readership of *Applied Physics Express*.

Sincerely,

Hiroki Yoshida,
Ryohei Kono,
Manato Fujimoto,
Motoki Asano,
Daiki Hatanaka,
Kei Yamamoto,
Shuichi Murakami